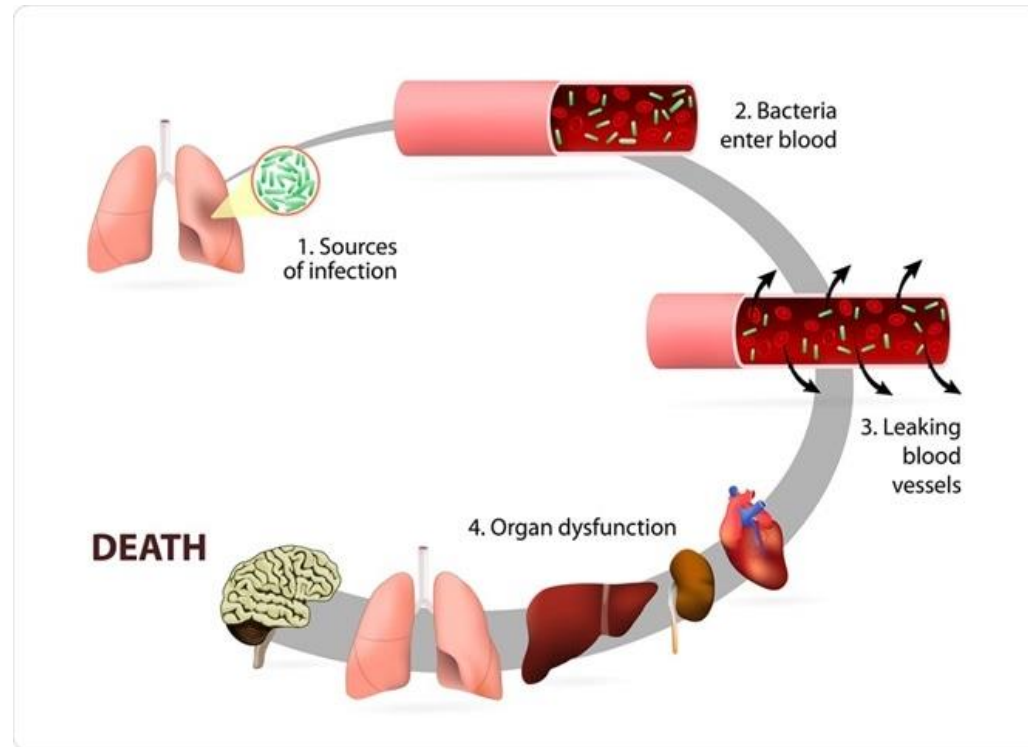


Identifying Dead-end States in Sepsis Patients With Reinforcement Learning

By: Charles Schepanowski, Eric Buitron Lopez, Keaton Banik and Mohammad Ahsan Siddiqui

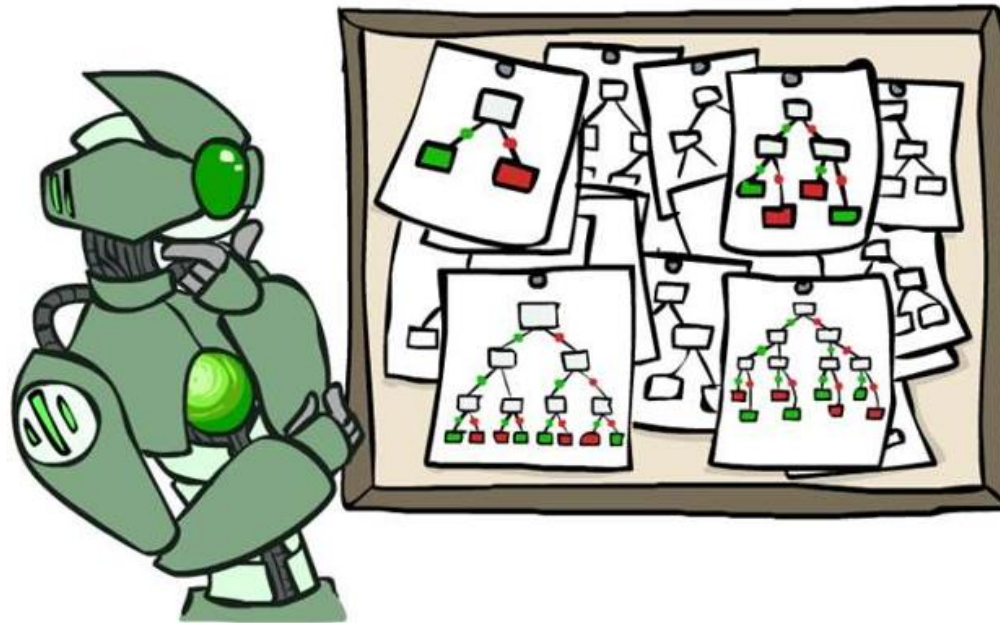
Background and Motivation

- Sepsis is a life-threatening condition caused by the body's extreme response to an infection.
- Sepsis leads to organ dysfunction and can be fatal if not treated promptly.
- A "medical dead-end" or "dead-end state" indicates a point where a patient will expire regardless of further treatment.



Problem Statement

- Our goal is to use reinforcement learning to identifying which treatments to avoid to prevent entering a dead-end state
- Identifying treatments to avoid is safer and more reliable than recommending optimal treatments based on uncertain RL estimates.

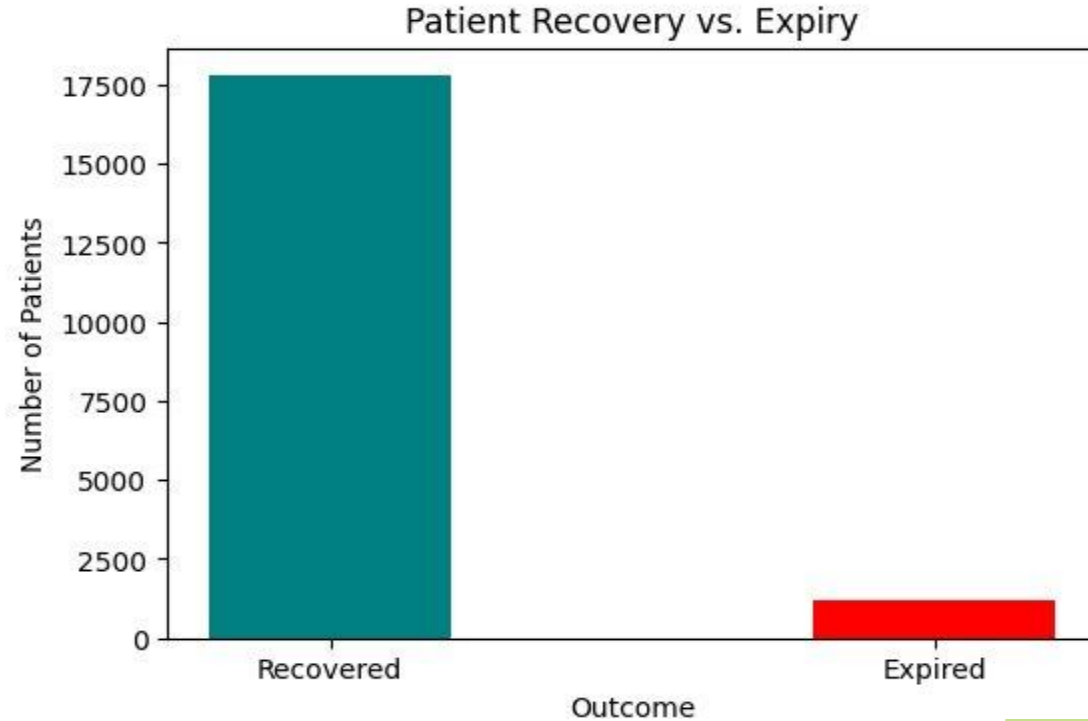


Methodology

- To frame as an RL problem, we define:
 - **States:** as a combination of multiple features (e.g., age, blood pressure) and are updated every 4 hours
 - **Actions:** 25 combinations (5 levels of IV fluids × 5 levels of vasopressors)
 - **Rewards:** as +1 for recovery, -1 for expiry, and 0 otherwise
- DQN was chosen for its ability to handle high-dimensional clinical data and learn from historical (off-policy) treatment decisions.
- Two separate DQNs were used to estimate death risk Q_D and recovery potential Q_R

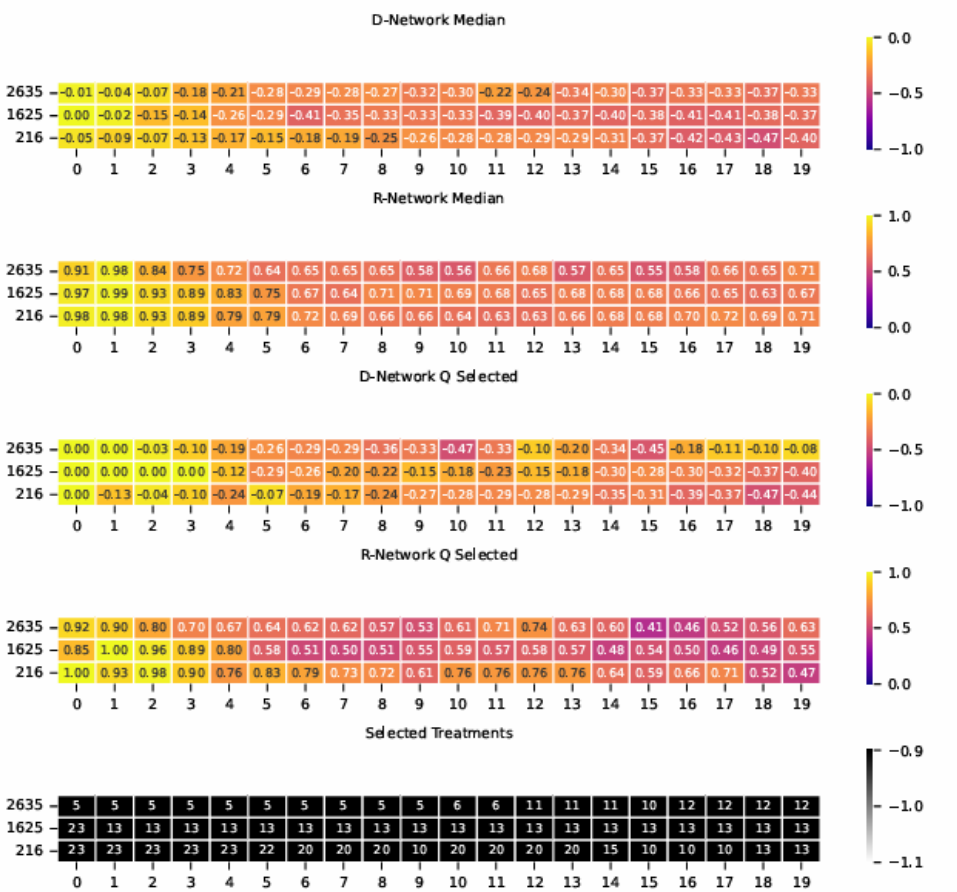
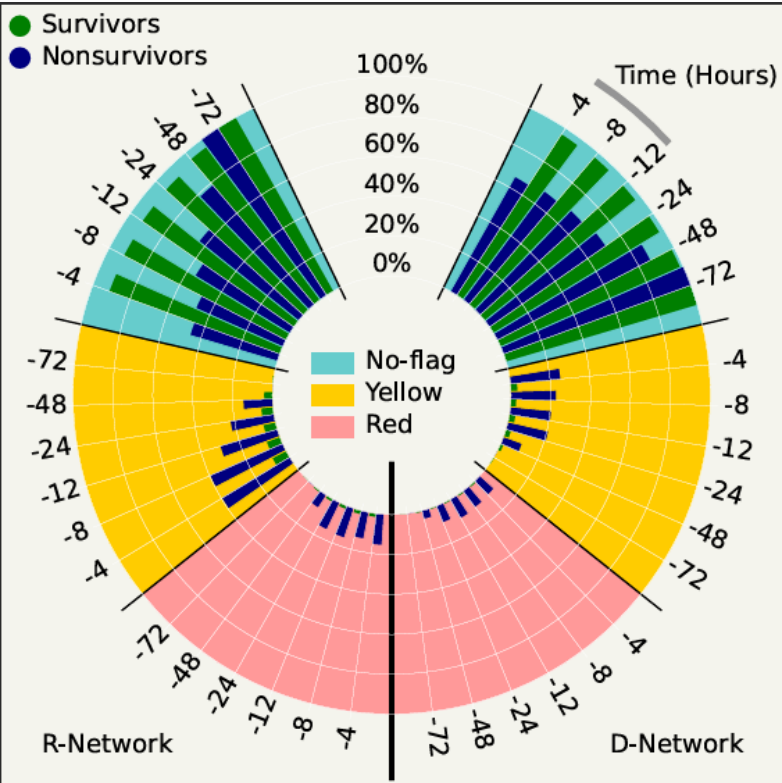
Data

- We use the MIMIC-III dataset which is a publicly available ICU dataset containing real patient records.
- The dataset includes age, vital signs, medications, medical interventions and more.
- A key limitation in the data is the imbalance of recovered vs expired patients.



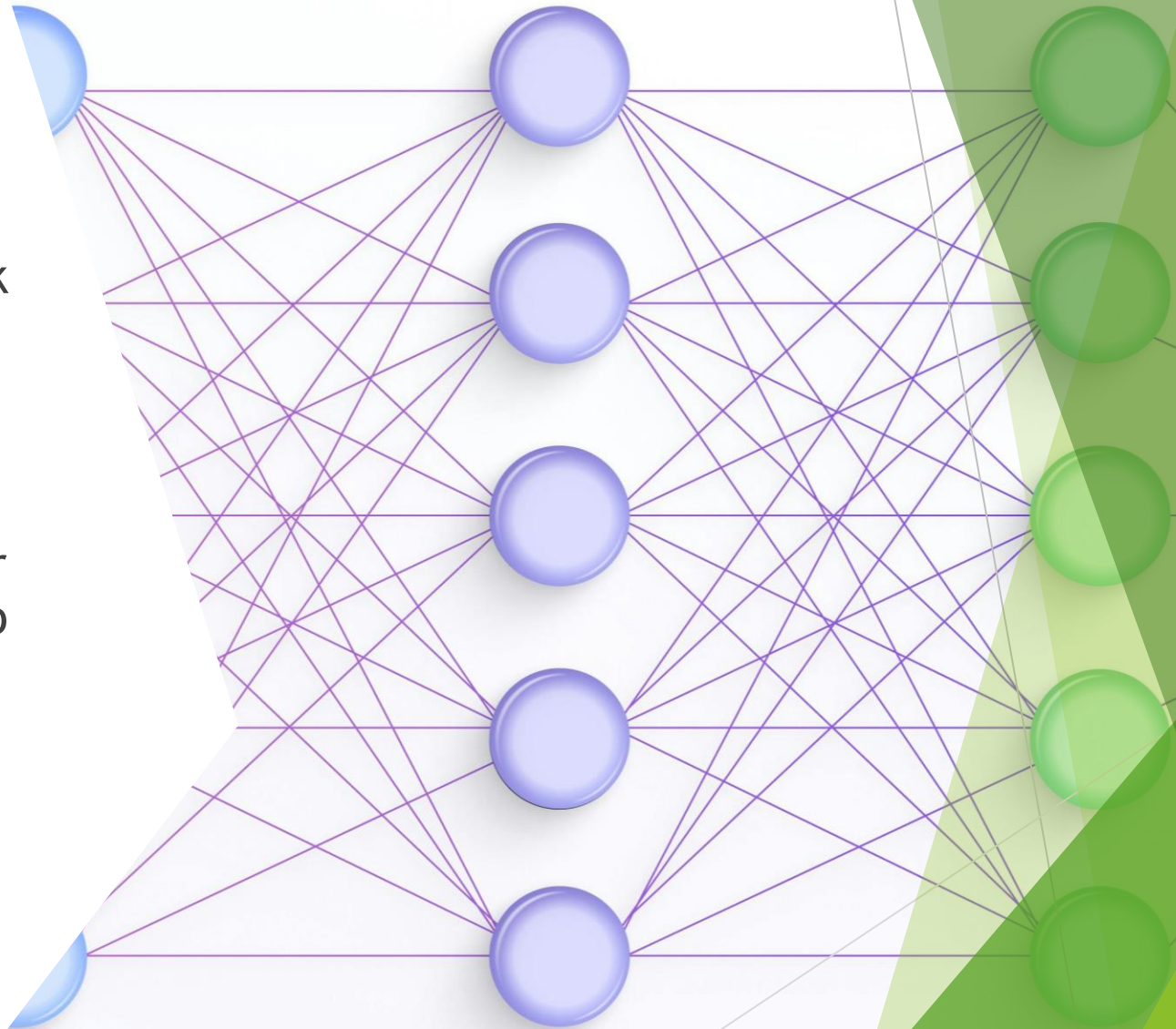
Preliminary Results

- Successfully extracted sepsis patient data from the MIMIC-III dataset.
- Reproduced key results from the original DeD paper using the provided codebase.
- Conducted initial experiments with minor modifications to the original DQN architectures.



Next Steps

- Experiment with different DQN architectures by varying network depth and layer sizes to identify the optimal configuration for predicting dead-end states.
- Replace DQN with alternative methods like Behavior Cloning or Conservative Q-learning (CQL) to compare performance.
- Explore alternative reward structures, such as introducing intermediate rewards to better reflect patient progress.



The background features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green, creating a modern, layered effect on the right side of the slide.

Thank you!

Questions?